Tarbiat Modares University
Faculty of Mechanical Engineering

TARBIAT MODARES UNIVERSITY
Faculty of Mechanical Engineering – Department of Mechanical Engineering

Certificate of Research Assistantship

To Whom It May Concern,

This is to certify that **Amir Abbas Khalili** served as a **Research Assistant** under my supervision in the Department of Mechanical Engineering, Tarbiat Modares University, from **July 23, 2026, to September 23, 2026**.

During this period, he contributed to research activities in computational mechanics, numerical modeling, and the simulation of advanced structures. His responsibilities and contributions included:

· Conducting a literature review and technical investigation related to wave propagation and numerical modeling of three-dimensional structures.
· Contributing to the development and validation of appropriate methodologies for determining and analyzing wave propagation paths in three-dimensional structures.
· Investigating and developing approaches for plasticity and failure modeling in Abaqus/Explicit, including the definition of the required plastic material behavior and the implementation of element deletion through keyword-level settings, enabling a more appropriate numerical representation of structural failure and material damage.
· Preparing and recording step-by-step instructional materials and video tutorials covering the developed modeling and simulation procedures in COMSOL Multiphysics and Abaqus.

Throughout this period, he demonstrated strong technical skills, research initiative, and commitment to the assigned research activities.

This certificate is issued at his request to confirm his research experience and contributions under my supervision.

Sincerely,

Dr. Morteza Karamooz Mahdiabadi
Assistant Professor of Mechanical Engineering
Department of Mechanical Engineering
Tarbiat Modares University

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Signature: _____

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